

MARKET INTELLIGENCE: REQUIREMENTS

Project: *U.S. National Science Foundation (NSF) and German Federal Agency for Breakthrough Innovation (SPRIND) Challenge Program*

1.0 Introduction:

In October 2024, the Directorate for Technology, Innovation and Partnerships (TIP) at the U.S. National Science Foundation (NSF) announced a partnership with the German Federal Agency for Breakthrough Innovation ([SPRIND](#)) to accelerate breakthrough innovations in key technology areas. NSF and SPRIND, collectively the “funding agencies,” jointly seek to enable researchers, innovators and entrepreneurs to apply unconventional approaches to create game-changing technologies and translate discoveries into tangible applications and products utilizing a process based on the SPRIND Challenge model.

NSF seeks the support of a contractor who can help attract proposals from applicants and entrepreneurs from every corner of the United States, host oral pitch competitions, significantly accelerate proposal processing timelines from the status quo, increase opportunities to support high-risk, high-reward ideas, support collaborations with international experts on emerging technologies, and provide mentoring and technical assistance to enable teams to maximize their potential. NSF is seeking input from interested parties across the nation that could design and implement a fast, accessible and nimble process to support and guide researchers, innovators and entrepreneurs to meet predefined milestones and objectives and translate discoveries into products and services. This program is intended to test an alternative approach to the NSF merit review process that significantly accelerates timelines to fund innovative and potentially disruptive translational research, all the while empowering TIP with data to inform how it makes investments in the future.

The purpose of this request for information is to conduct market intelligence for information gathering. Specifically, NSF is soliciting information from interested parties about the NSF and SPRIND Challenge Program as currently envisioned. Respondents are encouraged to offer their own insights on the activities that would best support the agency’s goals of enabling researchers and innovators to apply unconventional approaches to create game-changing technologies and translate discoveries into tangible applications and products utilizing a process based on the SPRIND Challenge model.

NSF is also seeking an organization that has the operational nimbleness, organizational capability and experience to attract proposals from first-time applicants and entrepreneurs from every corner of our country; create space for new thinkers from science and business to take risks and think radically differently; and support collaborations with international experts on emerging technologies. As speed matters in innovation, the NSF and SPRIND Challenge Program will follow the SPRIND model including [timelines typical of SPRIND Challenges](#). Specifically, NSF is looking for an organization that can execute the tasks and activities described under each of the four primary workstreams in Section 4.0 to accomplish the performance objectives contained in Section 5.0.

2.0 Period and Place of Performance:

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3.0 Instrument and Approach:

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4.0 Eligibility:

Any responsible source (excluding foreign entities) may submit proposals for consideration. All offerors must be registered in the Federal Government's System for Award Management (SAM) prior to submitting a proposal (www.sam.gov). A proposal can involve multiple organizations, but the proposal must be submitted by a lead organization with subcontracts to other participating organizations. An eligible organization can submit a maximum of one proposal as a lead.

Characteristics of a Strong Organization(s)

NSF seeks contractors with:

- An established network with expertise working with and capability to stimulate the translation of discoveries into tangible applications and products through coaching and subawards or subcontracts.
- Experience working in a highly collaborative environment with scientific funding agencies, U.S. federally-funded researchers and their labs, technology accelerators, and other stakeholders in the United States and Europe (especially Germany).
- Experience working with SPRIND or similar organizations which, like SPRIND, are focused on fostering groundbreaking, high-risk technological innovations with potential for significant societal or economic impact.
- Marketing and communications capacity as well as experience working with deep tech venture capital investors and industry.
- Organizational capability to respond quickly and nimbly to speed funding to the point of innovation, respond to changes and otherwise adapt as needed to achieve program goals.

5.0 Performance Objectives:

NSF will provide the topic to the selected contractor. The workstreams for support sought from the contractor can be organized into four major categories described here and, in the Roles and Responsibilities Chart included as part of this market intelligence.

Workstream 1: Outreach and cohort selection:

- The contractor will format the topic description, end goal and milestones detailed by NSF and SPRIND into a call for proposals and invite researchers, innovators and entrepreneurs to apply for funding.
- The contractor will organize communications and outreach activities through various channels (e.g., website, social media, traditional media, webinars) to raise awareness of the funding opportunity amongst researchers, innovators and entrepreneurs throughout the United States.
- The contractor will coordinate the proposal review process. The selection criteria for the proposals will be detailed by NSF and SPRIND, and key personnel from the funding

agencies will utilize these criteria to screen proposals and short-list the most promising proposals for further consideration. NSF, SPRIND and the contractor will work together to identify internationally-renowned technical and industry experts from various research fields and disciplines to serve on the “jury” (i.e., the review panel) to review proposals alongside representatives from NSF and SPRIND.

Workstream 2: Subcontracting with the selected teams:

- The contractor will be responsible for issuing awards to the U.S. teams and distributing funds within days of the selection.
- To the extent that it is possible and prudent, and with NSF guidance and assistance, the contractor will structure award conditions to relieve teams from the administrative burden to comply with complicated award terms and conditions so more of the teams’ time and resources are focused on performance and meeting the stated goals and milestones of the funding opportunity.
- The contractor will host and handle communications and logistics for the pitch competition, which may be hybrid – with the jury in person and the proposing teams joining virtually.
- Teams are initially supported for one year. As that year closes, the contractor will host the teams, jury members and key personnel from the funding agencies – ideally in person – so that the teams can be assessed by the jury based on progress towards the first-year milestone. A sub-set of the original cohort of awardees will be selected to receive funding for Stage 2 (a second year). The contractor will again coordinate a down selection process with jury and funding agencies as the second year closes.

Workstream 3: Mentoring and technical assistance for teams:

- The contractor will be responsible for providing networking opportunities, coaching and technical assistance to the funded teams throughout the program to accelerate the translation of their research to market and impact.
- The contractor may organize workshops or webinars focused on essential skills such as pitching to investors, regulatory compliance and commercialization strategies.
- A strong source organization will have access to coaches with:
 - Expertise in translation and technology markets within the focal areas of the proposed funding opportunity;
 - Demonstrated success supporting the development of early-stage, high-tech start-ups;
 - Experience coaching, mentoring and supporting researchers to accelerate the translation of research to the market and impact;
 - Experience working with entrepreneurs and universities in the United States and Europe and linkages to expert networks in both places;
 - Networks of potential industry partners and venture capitalists; and
 - Demonstrated success connecting entrepreneurs to business services support, legal advice, team building support, and technical and entrepreneurial expertise.

Workstream 4: Documentation and reporting of progress towards milestones and key learnings:

- The contractor will track team progress towards the program goals and milestones and keep the funding agencies, coaches and jurors (as appropriate) informed of progress, including hurdles or situations that may require a pivot.
- The contractor will be responsible for organizing data and documenting learnings, as NSF seeks to learn from the success and challenges of this program.
 - This includes developing and sharing with NSF any resources (e.g., templates, written procedures, coaching materials, guidelines or other tools/artifacts) that could help standardize or make the process repeatable should NSF elect to conduct future funding opportunities.

6.0 Performance Objectives/Project Deliverables:

Interested parties should consider the workstreams of support, which can be organized into four major categories. Key objectives are summarized below. Also refer to the Roles and Responsibilities Chart. All steps must allow for a maximally efficient and effective process.

1. Outreach and cohort selection:
 - a. Develop the call for proposals; a streamlined, user-friendly application process that is straightforward for all potential applicants, including those who have never received NSF funding; and a secure platform to facilitate efficient coordination with NSF, SPRIND and jury members.
 - b. Develop and publish a Participant Agreement that mirrors much of the SPRIND version, to the extent allowed by law, and puts minimal administrative burden on the teams. For example, refer to the sample agreement included as part of this market intelligence package.
 - c. Engage with researchers, innovators and entrepreneurs from various communities (i.e., universities, incubators/accelerators, small business support agencies, local and regional business support agencies, etc.) to promote this funding opportunity across the full geography of the United States.
 - d. Coordinate the proposal review process such that the time from proposal submission to award notification is approximately six weeks.
 - e. Coordinate the down-selection process for years two and three.
2. Subcontracting with the selected teams:
 - a. Notify selected teams.
 - b. Execute a Participant Agreement with U.S. teams. For example, refer to the sample agreement included as part of this market intelligence package.
 - c. Distribute funding. For example, 50% within five business days of award notification and additional funding increments (25%) within six months and nine months. Repeat for years two and three.
3. Mentoring and technical assistance for teams:
 - a. Provide biweekly individual coaching to awardees in areas of technical development and business development.
 - b. Provide cohort support by organizing and hosting quarterly, in-person events for awardees, coaches and funding agency staff.

- c. Facilitate networking and introductions to subject matter and business experts, investment groups, lawyers and experts in intellectual property protection, regulations in the United States and Europe, and techno-economic analysis.
- 4. Documentation and reporting of activities, progress towards milestones and key learnings
 - a. Report to NSF and SPRIND on progress towards milestones and goals at regular cadences and ad hoc, as needed.
 - b. Summarize key learnings and share templates, written procedures, coaching materials, guidelines or other tools/artifacts at regular cadences and ad hoc, as needed.

7.0 Market Intelligence Responses:

This market intelligence effort is an opportunity for NSF to hear from and receive feedback from potential proposers as well as the broader community regarding:

1. **Prospective Vision and Capabilities:** Please provide your own vision up to three pages to execute the program and achieve the performance objectives under each of the four primary workstreams. Include the specific capabilities, resources, experiences, networks and activities you would utilize to support this program, including (but not exclusively) demonstrated understanding of European State Aid Laws related to procurement and contracting. Specify which workstream(s), if any, would need to be subcontracted out. Highlight your vision and capabilities as it applies to teams of U.S. researchers and innovators as well as teams with participants from Germany and other international locations. Responses are optional but will help NSF gain a better understanding of the universe of organizations capable of and interested in supporting this and future technology translation activities.
2. **Feedback:** If you have feedback, questions, areas of concern or suggestions on how this program could be structured as outlined in this document, this is an opportunity to provide that feedback. Questions and feedback may be submitted directly in response to this market intelligence effort. Questions and feedback may be included at the end of the Prospective Vision and Capabilities document (and do not count towards the page limit) or separately. NSF is especially interested in a high-level breakdown of the estimated costs of the activities described and the suggested roles and responsibilities of the contractor in relation to NSF and SPRIND (refer to the Roles and Responsibilities Chart included in this market intelligence package and the relevant section of this document).

Companies and individuals are welcome to submit information regarding Prospective Vision and Capabilities or Feedback, or both.

This posting shall not be construed as an obligation on the part of the Government. This is NOT a request for quotations or proposals. No solicitation document exists, and a formal solicitation may or may not be issued by the Government as a result of the responses received to this posting. The Government will not pay for any response or demonstration expenses. All costs incurred

responding to this market intelligence will be solely at the interested party's expense. Failure to respond to this market intelligence will not preclude participation in any future solicitation.

Any information received becomes the property of the Government and will not be returned to the submitter. Interested parties are responsible for adequately marking proprietary or program sensitive information contained in their response. As desired, interested parties may also specify any restrictions on how their response may be shared or disclosed.

Submission Details: Please submit your responses to this market intelligence via email to Alexandra Roig Vitali at avitali@nsf.gov before 11 April 2025, at 12:00 PM Eastern Time. When submitting your response, please ensure that attachments do not exceed 20 MB in size and reference NSF-SPRIND-FY25-RFI in the email subject.

8.0 Agency Contacts:

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